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Comprehensive Question Bank: Chapter 2 — Computational Thinking & Algorithms

Section A: Multiple-Choice Questions (MCQs)

Topic 2.1: Introduction to Computational Thinking

1. Computational thinking is best defined as:
A) A method used only by professional programmers
B) A way of thinking to solve problems clearly and logically
C) A type of programming language
D) A hardware component of a computer
2. Which of the following is NOT one of the four core habits of computational thinking?
A) Decomposition
B) Pattern Recognition
C) Memorization
D) Abstraction
3. In computational thinking, "abstraction" refers to:
A) Breaking a problem into smaller parts
B) Ignoring unimportant details and focusing on what matters
C) Creating step-by-step instructions
D) Testing a solution for errors
4. Which of the following is an example of pattern recognition in daily life?
A) Planning a trip by breaking it into smaller tasks
B) Sorting books by subject or size
C) Writing a step-by-step recipe
D) Testing a solution to see if it works
5. Which of the following best illustrates computational thinking being used outside of computer science?
A) Only software engineers writing code
B) A chef planning a complex, multi-course meal step by step
C) A computer executing binary instructions
D) A router forwarding data packets

6. Making a daily timetable is an example of using computational thinking because it involves:
- A) Random guessing
 - B) Organizing tasks in a clear, logical order**
 - C) Writing computer code
 - D) Avoiding all planning
7. Which statement about computational thinking is TRUE?
- A) It can only be applied to mathematical problems
 - B) It can be applied to daily life activities as well as computer science problems**
 - C) It is only useful for people who already know how to code
 - D) It replaces the need for algorithms
-

Topic 2.2: Problem Solving Using Computational Thinking

1. What is the first and most important step in problem solving using computational thinking?
- A) Writing code
 - B) Identifying the problem clearly**
 - C) Evaluating the solution
 - D) Breaking the problem into parts
2. Breaking a large problem into smaller, manageable parts is known as:
- A) Abstraction
 - B) Decomposition**
 - C) Evaluation
 - D) Pattern recognition
3. A step-by-step solution used to solve a problem is known as a/an:
- A) Program only
 - B) Flowchart only
 - C) Algorithm**
 - D) Proposition
4. Why is decomposition useful when solving a large problem?
- A) It makes the problem more confusing
 - B) It allows each smaller part to be solved individually, reducing confusion and stress**
 - C) It eliminates the need for a solution
 - D) It combines multiple problems into one
5. What is the purpose of evaluating a solution after it has been created?
- A) To make the solution more complicated
 - B) To check whether the solution is correct and to find ways to improve it**
 - C) To skip the testing phase entirely
 - D) To immediately discard the solution

6. Which of the following best describes the correct order of the problem-solving stages taught in this chapter?
- A) Evaluate → Decompose → Identify → Create steps
 - B) Identify the problem → Decompose → Create step-by-step solution → Evaluate and improve**
 - C) Create steps → Identify → Evaluate → Decompose
 - D) Decompose → Evaluate → Identify → Create steps
7. A key benefit of clearly identifying a problem before solving it is:
- A) It makes the problem harder to understand
 - B) It saves time and effort by focusing only on relevant information**
 - C) It removes the need for an algorithm
 - D) It guarantees the solution will be perfect on the first attempt
-

Topic 2.3: Logic as a Knowledge Representation Framework

1. In computer science, logic is primarily used to:
- A) Increase the physical speed of hardware
 - B) Represent facts and rules so correct, consistent decisions can be made**
 - C) Replace the need for algorithms entirely
 - D) Store large amounts of unstructured data
2. Logical statements in programs are primarily used to:
- A) Store images and videos
 - B) Control the flow of programs based on true/false conditions**
 - C) Increase Internet bandwidth
 - D) Physically connect hardware devices
3. Which of the following fields commonly relies on logic for reasoning and decision-making?
- A) Only video editing software
 - B) Expert systems and artificial intelligence**
 - C) Only physical cable manufacturing
 - D) Only printer hardware design
4. An "if-then" statement such as "If it rains, then we will stay indoors" is an example of:
- A) A random guess
 - B) A logical rule used for reasoning and decision-making**
 - C) A hardware instruction
 - D) A network protocol

5. What is the main benefit of using logic instead of guesswork in decision-making?
- A) It makes decisions slower
 - B) It removes confusion and ensures decisions are consistent and accurate**
 - C) It eliminates the need for facts
 - D) It only works for simple problems
-

Topic 2.4: Propositional Logic

1. A proposition is best defined as:
- A) A question that requires an answer
 - B) A statement that is either true or false, but not both**
 - C) A command given to a computer
 - D) Any sentence in the English language
2. Which of the following is a proposition?
- A) "Close the door!"
 - B) "What time is it?"
 - C) "The sun rises in the east."**
 - D) "Please help me."
3. Which of the following is NOT a proposition?
- A) "Today is Sunday."
 - B) "Is it raining?"**
 - C) "The sky is blue."
 - D) " $2 + 2 = 4$."
4. A simple proposition is best described as:
- A) A statement formed by combining two or more statements
 - B) A single statement that cannot be broken into smaller parts**
 - C) A statement that is always false
 - D) A question with no clear answer
5. A compound proposition is formed by:
- A) Using only one simple statement
 - B) Combining two or more simple propositions using logical connectives**
 - C) Removing all logical connectives
 - D) Asking a question
6. Which symbol represents the logical connective AND?
- A) $\vee$
 - B) $\wedge$**
 - C) $\neg$
 - D) $\leftrightarrow$

7. Which symbol represents the logical connective OR?
- A) $\wedge$
 - B) $\vee$**
 - C) $\neg$
 - D) $\rightarrow$
8. Which symbol represents the logical connective NOT?
- A) $\wedge$
 - B) $\vee$
 - C) $\neg$**
 - D) $\leftrightarrow$
9. Which symbol represents the IMPLIES (conditional) connective?
- A) $\wedge$
 - B) $\vee$
 - C) $\rightarrow$**
 - D) $\neg$
10. Which symbol represents the BICONDITIONAL connective?
- A) $\rightarrow$
 - B) $\vee$
 - C) $\leftrightarrow$**
 - D) $\neg$
11. The expression $P \wedge Q$ is true only when:
- A) At least one of P or Q is true
 - B) Both P and Q are true**
 - C) P is false
 - D) Q is false
12. The expression $P \vee Q$ is true when:
- A) Both P and Q must be true
 - B) At least one of P or Q is true**
 - C) Neither P nor Q is true
 - D) Only P is true and Q is false
13. The expression $\neg P$ is true when:
- A) P is true
 - B) P is false**
 - C) P is either true or false
 - D) P is a compound proposition

14. In the conditional statement $P \rightarrow Q$, the statement is considered false only when:
- A) P is false and Q is false
 - B) P is false and Q is true
 - C) P is true and Q is false**
 - D) P is true and Q is true
15. The biconditional statement $P \leftrightarrow Q$ is true when:
- A) P and Q always have different truth values
 - B) P and Q have the same truth value**
 - C) P is always true
 - D) Q is always false
16. Which real-world example correctly models a compound proposition using AND and OR, similar to a streaming platform login check?
- A) "Enter your username."
 - B) "User has correct password AND (has active subscription OR is on free trial)."**
 - C) "Click the login button."
 - D) "Forgot your password?"
-

Topic 2.5: Truth Tables

1. What is the main purpose of a truth table?
- A) To store user data
 - B) To show the truth value of a logical expression for every possible combination of inputs**
 - C) To design computer hardware
 - D) To write program code
2. How many rows will a truth table have for an expression with 2 propositions?
- A) 2
 - B) 3
 - C) 4**
 - D) 8
3. How many rows will a truth table have for an expression with 3 propositions?
- A) 4
 - B) 6
 - C) 8**
 - D) 9

4. The general formula for the number of rows in a truth table with n propositions is:
- A) n^2
 - B) 2^n**
 - C) $n + 2$
 - D) $2 \times n$
5. In the truth table for $P \wedge Q$, how many rows result in a TRUE outcome?
- A) 0
 - B) 1**
 - C) 2
 - D) 4
6. In the truth table for $P \vee Q$, how many rows result in a TRUE outcome?
- A) 1
 - B) 2
 - C) 3**
 - D) 4
7. According to the truth table for $P \rightarrow Q$, when P is False, the result is:
- A) Always False
 - B) Always True**
 - C) Equal to Q 's value only
 - D) Undefined
8. What is a tautology?
- A) An expression that is always False
 - B) An expression that is always True, regardless of input values**
 - C) An expression with no truth value
 - D) An expression that cannot be tested
9. What is a contradiction?
- A) An expression that is always True
 - B) An expression that is always False, regardless of input values**
 - C) An expression with exactly one true row
 - D) An expression that cannot have a truth table
10. What is a contingency?
- A) An expression that is always True
 - B) An expression that is always False
 - C) An expression that is True in some cases and False in others**
 - D) An expression with no propositions

11. The first step in creating a truth table is to:

- A) Apply logical connectives
 - B) Identify all the propositions involved**
 - C) Write the final answer
 - D) Simplify the expression
-

Topic 2.6: Propositional Equivalence

1. Propositional equivalence means that two logical expressions:

- A) Look exactly the same when written
- B) Always produce the same truth values in every possible case**
- C) Have a different number of propositions
- D) Can never be compared

2. Which symbol is used to represent logical equivalence?

- A) $\rightarrow$
- B) $\equiv$**
- C) $\neg$
- D) $\vee$

3. How can two logical expressions be proven equivalent?

- A) By guessing
- B) By comparing their truth tables row by row and confirming the outputs match**
- C) By counting the number of variables
- D) By checking if they use the same connectives

4. According to De Morgan's Law, $\neg(P \wedge Q)$ is equivalent to:

- A) $\neg P \wedge \neg Q$
- B) $\neg P \vee \neg Q$**
- C) $P \vee Q$
- D) $P \wedge Q$

5. According to De Morgan's Law, $\neg(P \vee Q)$ is equivalent to:

- A) $\neg P \vee \neg Q$
- B) $\neg P \wedge \neg Q$**
- C) $P \wedge Q$
- D) $P \rightarrow Q$

6. Who is credited with discovering the equivalence laws that allow negated AND/OR statements to be transformed into each other?

- A) George Boole

B) Alan Turing

C) Augustus De Morgan

D) Aristotle

7. Why is propositional equivalence important in programming?

A) It makes code slower

B) It allows complex expressions to be simplified without changing their meaning, improving efficiency

C) It eliminates the need for logic entirely

D) It only applies to hardware design

8. The expression $P \vee \neg P$ is an example of a:

A) Contradiction

B) Contingency

C) Tautology

D) Predicate

9. The expression $P \wedge \neg P$ is an example of a:

A) Tautology

B) Contradiction

C) Contingency

D) Quantifier

Topic 2.7: Propositional Satisfiability

1. A proposition is considered satisfiable if:

A) It is always false

B) There is at least one combination of truth values that makes it true

C) It contains no propositions

D) It cannot be tested using a truth table

2. A proposition is considered unsatisfiable if:

A) It is true in at least one case

B) Every possible combination of truth values makes it false

C) It has more than two variables

D) It uses the OR connective

3. Which of the following expressions is satisfiable?

A) $P \wedge \neg P$

B) $P \vee Q$

C) An expression with no propositions

D) A statement that is always false

4. Which of the following expressions is unsatisfiable?
- A) $P \vee Q$
 - B) $P \wedge Q$
 - C) $P \wedge \neg P$**
 - D) $\neg P \vee Q$
5. What is the "SAT problem" in computer science?
- A) A method for designing network topologies
 - B) The problem of determining whether a logical expression can be made true by some assignment of values**
 - C) A type of hardware chip
 - D) A method for compressing files
6. Real-world SAT solvers are commonly used for:
- A) Increasing Internet bandwidth
 - B) Verifying software correctness and checking safety-critical systems like autopilot software**
 - C) Designing physical network cables
 - D) Formatting hard drives
7. In the expression "Bring pizza OR (burger AND do NOT bring soda)," which outcome makes the expression satisfiable?
- A) No solution exists
 - B) Bringing pizza and no soda**
 - C) The expression can never be true
 - D) Bringing only soda
8. The expression "You must bring pizza AND you must NOT bring pizza" is:
- A) Satisfiable
 - B) Unsatisfiable**
 - C) A tautology
 - D) A predicate
-

Topic 2.8: Predicate Logic

1. Predicate logic is best described as:
- A) A simpler version of propositional logic
 - B) An extension of propositional logic that allows statements about objects, properties, and relationships**
 - C) A type of network topology
 - D) A method for creating truth tables only

2. A predicate is best defined as:
- A) A statement that is always true
 - B) A statement whose truth value depends on the value of one or more variables**
 - C) A fixed proposition with no variables
 - D) A type of logical connective
3. Which of the following is an example of a predicate?
- A) "Today is Sunday."
 - B) "IsEven(x)"**
 - C) "The sky is blue."
 - D) " $P \wedge Q$ "
4. What is the key limitation of propositional logic that predicate logic overcomes?
- A) Propositional logic cannot use the AND connective
 - B) Propositional logic cannot look inside a statement to separate the subject from its property**
 - C) Propositional logic cannot be written using symbols
 - D) Propositional logic has no truth values
5. In the example " $A = \text{'Ali is Tall'}$ " versus " $\text{Tall}(\text{Ali})$," which statement is written in predicate logic?
- A) $A = \text{'Ali is Tall'}$
 - B) $\text{Tall}(\text{Ali})$**
 - C) Both are predicate logic
 - D) Neither is predicate logic
6. In predicate logic, a variable such as x represents:
- A) A fixed truth value
 - B) An unspecified object from a domain of possible values**
 - C) A logical connective
 - D) A truth table row
7. Compared to propositional logic, predicate logic is:
- A) Less expressive and less flexible
 - B) More expressive and more powerful**
 - C) Identical in expressive power
 - D) Unable to represent relationships
8. Which of the following correctly represents "student x borrows book y " in predicate logic?
- A) $\text{Student}(x) \wedge \text{Book}(y)$
 - B) $\text{Borrows}(x, y)$**
 - C) $x \wedge y$
 - D) $\neg \text{Borrows}(x, y)$
-

Topic 2.9: Quantifiers in Predicate Logic

1. What is the purpose of quantifiers in predicate logic?
A) To replace all predicates with propositions
B) To describe how many objects in a domain satisfy a given condition
C) To remove variables from statements
D) To convert predicate logic into propositional logic
2. Which symbol represents the universal quantifier?
A) $\exists$
B) $\forall$
C) $\wedge$
D) $\neg$
3. Which symbol represents the existential quantifier?
A) $\forall$
B) $\exists$
C) $\rightarrow$
D) $\leftrightarrow$
4. The statement " $\forall x, x > 0$ " means:
A) There exists one x greater than 0
B) For all x , x is greater than 0
C) No x is greater than 0
D) x is sometimes greater than 0
5. The statement " $\exists x, x$ is even" means:
A) All values of x are even
B) There exists at least one x that is even
C) No value of x is even
D) x is never even
6. Which quantifier would be used to express the statement "All students are hardworking"?
A) Existential quantifier ($\exists$)
B) Universal quantifier ($\forall$)
C) Neither quantifier is needed
D) Both quantifiers must be used together
7. Which quantifier would be used to express the statement "Some books are interesting"?
A) Universal quantifier ($\forall$)
B) Existential quantifier ($\exists$)
C) Neither quantifier is needed
D) A logical connective, not a quantifier

8. What is the meaning of the statement " $\forall x \exists y, \text{Loves}(x, y)$ "?
 - A) There is one specific person loved by everyone
 - B) Every person loves at least someone (not necessarily the same person)**
 - C) No one loves anyone
 - D) Everyone loves everyone equally
 9. What is the meaning of the statement " $\exists y \forall x, \text{Loves}(x, y)$ "?
 - A) Every person loves at least someone, possibly different for each person
 - B) There exists one specific person who is loved by everyone**
 - C) No one loves anyone
 - D) The statement has no valid meaning
 10. Which of the following is TRUE regarding quantifier order?
 - A) The order of $\forall$ and $\exists$ never changes the meaning of a statement
 - B) Changing the order of $\forall$ and $\exists$ can completely change the meaning of a statement**
 - C) Only $\forall$ can be used in real statements
 - D) Only $\exists$ can be combined with predicates
 11. Does " $\exists y \forall x, \text{Loves}(x, y)$ " being true guarantee that " $\forall x \exists y, \text{Loves}(x, y)$ " is also true?
 - A) No, they are unrelated statements
 - B) Yes, if one specific person is loved by everyone, then everyone loves someone**
 - C) Only if x and y represent the same object
 - D) Only in propositional logic, not predicate logic
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Topic 2.10: Applying Predicate Logic to Real-World Problems

1. In the library system example, which predicate correctly represents "teacher t teaches student s "?
 - A) $\text{Student}(t, s)$
 - B) $\text{Teaches}(t, s)$**
 - C) $\text{Borrows}(t, s)$
 - D) $\text{Available}(t, s)$
2. In the library system example, which predicate correctly represents "student x borrows book y "?
 - A) $\text{Teaches}(x, y)$
 - B) $\text{Borrows}(x, y)$**
 - C) $\text{Student}(x, y)$
 - D) $\text{Available}(x, y)$

3. The statement " $\forall x, \text{Student}(x) \rightarrow \text{Borrows}(x, \text{Book})$ " means:
- A) No student borrows a book
 - B) Every student borrows a book**
 - C) Some students borrow a book
 - D) Only one student borrows a book
4. The statement " $\exists y, \text{Book}(y) \wedge \text{Available}(y)$ " means:
- A) All books are available
 - B) There exists at least one book that is available**
 - C) No books are available
 - D) Every book is unavailable
5. Given the rule " $\forall x, \text{Student}(x) \rightarrow \text{HasLibraryCard}(x)$ " and the fact " $\text{Student}(\text{Ali})$," what can be logically concluded?
- A) Ali does not have a library card
 - B) Ali has a library card**
 - C) No conclusion can be drawn
 - D) Ali is not a student
6. Predicate logic can be used to model which of the following real-world systems?
- A) Only computer programs
 - B) Family trees, social networks, and computer programs**
 - C) Only physical hardware
 - D) Only mathematical equations with no real-world use
7. Applying predicate logic to real-world problems primarily helps us:
- A) Avoid using variables
 - B) Represent knowledge clearly and systematically to analyze problems and draw conclusions**
 - C) Eliminate the need for quantifiers
 - D) Replace propositional logic entirely with hardware logic
-

Topic 2.11: Logic-Based Reasoning and Inference

1. Logical inference is best defined as:
- A) Randomly guessing an answer
 - B) The process of deriving new facts from known information using logical rules**
 - C) Writing a truth table
 - D) Creating a predicate with no variables

2. Deduction is a reasoning method that:
 - A) Moves from a specific example to a general rule
 - B) Moves from general facts/rules to reach a specific conclusion**
 - C) Never guarantees a correct conclusion
 - D) Cannot be used by computers
3. In the argument "All humans are mortal. Socrates is a human. Therefore, Socrates is mortal," which type of reasoning is used?
 - A) Random guessing
 - B) Deduction**
 - C) Satisfiability testing
 - D) Decomposition
4. Which of the following correctly represents the Modus Ponens reasoning pattern?
 - A) $P \rightarrow Q, \neg Q, \text{ therefore } \neg P$
 - B) $P \rightarrow Q, P, \text{ therefore } Q$**
 - C) $P \rightarrow Q, Q \rightarrow R, \text{ therefore } P \rightarrow R$
 - D) $P \vee Q, \neg P, \text{ therefore } Q$
5. Which of the following correctly represents the Modus Tollens reasoning pattern?
 - A) $P \rightarrow Q, P, \text{ therefore } Q$
 - B) $P \rightarrow Q, \neg Q, \text{ therefore } \neg P$**
 - C) $P \wedge Q, \text{ therefore } P$
 - D) $P \vee Q, \text{ therefore } P$
6. In a smart home security system with the rule "IF motion is detected at night, THEN sound the alarm," and the fact "motion was detected at night," which reasoning pattern concludes "sound the alarm"?
 - A) Modus Tollens
 - B) Modus Ponens**
 - C) Syllogism only
 - D) Satisfiability testing
7. If a system rule states "If correct password entered (P), then access granted (Q)," and a user was NOT granted access ($\neg Q$), which reasoning pattern allows us to conclude the correct password was NOT entered ($\neg P$)?
 - A) Modus Ponens
 - B) Modus Tollens**
 - C) Decomposition
 - D) Abstraction

8. A syllogism combining " $P \rightarrow Q$ " and " $Q \rightarrow R$ " allows us to conclude:
- A) $P \wedge R$
 - B) $P \rightarrow R$**
 - C) $\neg P \rightarrow \neg R$
 - D) $Q \rightarrow P$
9. Why is deduction considered reliable in computer science and logic?
- A) It is based on guessing
 - B) The conclusion is guaranteed to be true if the premises are true and the reasoning is valid**
 - C) It only works for propositional logic, not predicate logic
 - D) It never requires any premises
10. Logical inference and deduction are especially important in which of the following fields?
- A) Only physical hardware manufacturing
 - B) Mathematics, computer science, and artificial intelligence**
 - C) Only network cabling
 - D) Only graphic design
-

Section B: Short Answer Questions (SQs)

Topic 2.1: Introduction to Computational Thinking

1. Define computational thinking in your own words.
2. List the four core habits of computational thinking.
3. Explain what is meant by "abstraction" in computational thinking.
4. Differentiate between decomposition and pattern recognition.
5. State three real-life examples of computational thinking mentioned in the chapter.
6. Explain why computational thinking is not limited to computer science students.
7. Identify which computational thinking habit is used when sorting books by subject or size.
8. Explain how planning a trip demonstrates computational thinking.
9. Predict which computational thinking habits you would use to plan a weekly study schedule.

Topic 2.2: Problem Solving Using Computational Thinking

1. What is meant by "identifying a problem clearly," and why is it the first step in problem solving?
2. Define decomposition in the context of problem solving.
3. Explain why decomposition reduces confusion and stress when solving a large problem.
4. Define an algorithm.

5. Explain why the steps of an algorithm must be in the correct order.
6. What is meant by "evaluating and improving solutions"?
7. Differentiate between creating a solution and evaluating a solution.
8. List, in correct order, the four stages of problem solving using computational thinking.
9. Explain how ignoring unnecessary details helps when identifying a problem.
10. Predict what might go wrong if a problem is not clearly identified before attempting a solution.

Topic 2.3: Logic as a Knowledge Representation Framework

1. Define logic as used in computer science.
2. Explain the role of logic in reasoning and decision making.
3. Explain how logical statements are used to control the flow of a computer program.
4. Give an example of an "if-then" statement and explain what it represents.
5. Explain why logic is important in expert systems and artificial intelligence.
6. Differentiate between decision-making based on logic and decision-making based on guesswork.
7. Explain how logic helps computers process information correctly.

Topic 2.4: Propositional Logic

1. Define a proposition.
2. Explain why questions and commands are not considered propositions.
3. Differentiate between a simple proposition and a compound proposition.
4. List the five logical connectives discussed in this chapter, along with their symbols.
5. Explain the meaning of the AND ($\wedge$) connective with an example.
6. Explain the meaning of the OR ($\vee$) connective with an example.
7. Explain the meaning of the NOT ($\neg$) connective with an example.
8. Explain the meaning of the IMPLIES ($\rightarrow$) connective with an example.
9. Explain the meaning of the BICONDITIONAL ($\leftrightarrow$) connective with an example.
10. Differentiate between the AND connective and the OR connective.
11. Translate the statement "It is raining and it is cold" into a compound proposition using a logical connective.
12. Identify the propositions and connective(s) in the statement: "You get free shipping if your order is over \$50 and you are a Premium Member."
13. Explain why $P \rightarrow Q$ is considered true when P is false, regardless of Q's value.

Topic 2.5: Truth Tables

1. Define a truth table.

2. Explain the purpose of a truth table in propositional logic.
3. State the formula used to calculate the number of rows in a truth table for n propositions.
4. Calculate how many rows a truth table would have for an expression with 3 propositions.
5. List the steps involved in creating a truth table for a logical expression.
6. Differentiate between a tautology, a contradiction, and a contingency.
7. Give an example of a tautology and explain why it is always true.
8. Give an example of a contradiction and explain why it is always false.
9. Predict the output of $P \wedge Q$ when P is True and Q is False.
10. Predict the output of $P \vee Q$ when both P and Q are False.
11. Predict the output of $P \rightarrow Q$ when P is False and Q is True.

Topic 2.6: Propositional Equivalence

1. Define propositional equivalence.
2. Explain how truth tables are used to identify equivalent logical expressions.
3. State De Morgan's first law in symbolic form.
4. State De Morgan's second law in symbolic form.
5. Explain why propositional equivalence is important in programming.
6. Differentiate between a tautology and an equivalence relationship between two expressions.
7. Explain, in your own words, why $\neg(P \wedge Q)$ is equivalent to $\neg P \vee \neg Q$.
8. Identify who is credited with the discovery of the logical laws relating negated AND/OR statements.
9. Explain how equivalence can help simplify a complex logical expression.

Topic 2.7: Propositional Satisfiability

1. Define propositional satisfiability.
2. Explain the condition under which a proposition is considered satisfiable.
3. Explain the condition under which a proposition is considered unsatisfiable.
4. Give an example of a satisfiable expression and explain why it is satisfiable.
5. Give an example of an unsatisfiable expression and explain why it is unsatisfiable.
6. Differentiate between a satisfiable expression and a tautology.
7. Explain what the "SAT problem" refers to in computer science.
8. Identify one real-world application of SAT solvers.
9. Predict whether the expression " $P \wedge \neg P$ " is satisfiable or unsatisfiable, and justify your answer.

Topic 2.8: Predicate Logic

1. Define predicate logic.
2. Define a predicate, and give one example.
3. Differentiate between propositional logic and predicate logic.
4. Explain why propositional logic cannot express a statement like "All students are hardworking," while predicate logic can.
5. Define what a variable represents in predicate logic.
6. Explain the difference between " $A = \text{'Ali is Tall'}$ " (propositional logic) and " $\text{Tall}(\text{Ali})$ " (predicate logic).
7. Translate the statement " x is greater than 5" into predicate logic notation.
8. Translate the statement "The teacher teaches the student" into predicate logic notation.
9. Explain why predicate logic is considered more expressive than propositional logic.

Topic 2.9: Quantifiers in Predicate Logic

1. Define the term "quantifier" in predicate logic.
2. Define the universal quantifier and state its symbol.
3. Define the existential quantifier and state its symbol.
4. Differentiate between the universal quantifier and the existential quantifier.
5. Translate the statement "All students are hardworking" using the universal quantifier.
6. Translate the statement "Some books are interesting" using the existential quantifier.
7. Explain, with an example, why the order of quantifiers in a statement can change its meaning.
8. Explain the difference in meaning between " $\forall x \exists y, \text{Loves}(x, y)$ " and " $\exists y \forall x, \text{Loves}(x, y)$."
9. Predict what " $\forall x, x > 0$ " means in plain English.
10. Predict what " $\exists x, x$ is even" means in plain English.

Topic 2.10: Applying Predicate Logic to Real-World Problems

1. Explain how predicates can be used to represent relationships between real-world objects.
2. Define the predicate " $\text{Borrows}(x, y)$ " and explain what it represents.
3. Define the predicate " $\text{Teaches}(t, s)$ " and explain what it represents.
4. Explain what the statement " $\forall x, \text{Student}(x) \rightarrow \text{Borrows}(x, \text{Book})$ " means.
5. Explain what the statement " $\exists y, \text{Book}(y) \wedge \text{Available}(y)$ " means.
6. Given the rule " $\forall x, \text{Student}(x) \rightarrow \text{HasLibraryCard}(x)$ " and the fact " $\text{Student}(\text{Ali})$," state the conclusion that can be drawn.
7. Explain how predicate logic can be applied to model a family tree or social network.

8. Identify two real-world systems, other than a library, that could be modeled using predicate logic.

Topic 2.11: Logic-Based Reasoning and Inference

1. Define logical inference.
 2. Define deduction as a reasoning method.
 3. Explain the Modus Ponens reasoning pattern with an example.
 4. Explain the Modus Tollens reasoning pattern with an example.
 5. Differentiate between Modus Ponens and Modus Tollens.
 6. Explain what a syllogism is, using the rules " $P \rightarrow Q$ " and " $Q \rightarrow R$."
 7. Explain why deduction is considered a reliable method of reasoning.
 8. Give a real-world example of Modus Ponens being used in a smart home or security system.
 9. Predict the conclusion that would result from applying Modus Tollens to the rule "If correct password entered, then access granted" combined with the fact "access was NOT granted."
 10. Explain why logical inference is important in fields like artificial intelligence and computer science.
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Section C: Long / Extensive Questions (LQs)

Topic 2.1: Introduction to Computational Thinking

1. Discuss computational thinking in detail as a universal problem-solving approach.
 - a) Define computational thinking and explain its four core habits: decomposition, pattern recognition, abstraction, and algorithm design.
 - b) Discuss at least five real-life examples of computational thinking mentioned in the chapter, explaining how each demonstrates one or more of the four habits.
 - c) Evaluate why computational thinking is considered useful for professionals outside of computer science, such as chefs, scientists, and engineers.
 2. Analyze how the four habits of computational thinking (decomposition, pattern recognition, abstraction, and algorithm design) could be applied together to plan a large real-world event, such as a school festival. Construct a step-by-step breakdown showing each habit in action.
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Topic 2.2: Problem Solving Using Computational Thinking

1. Discuss the four stages of problem solving using computational thinking in comprehensive detail.
 - a) Explain the importance of clearly identifying a problem before attempting to solve it.

- b) Describe decomposition in detail, including why it reduces confusion when solving large problems.
 - c) Discuss the role of creating step-by-step solutions (algorithms) and evaluating/improving those solutions.
2. Construct a full worked example applying all four problem-solving stages to a real-world technical problem (e.g., "a school's Wi-Fi keeps disconnecting"). Trace through identifying the problem, decomposing it into possible causes, building a step-by-step diagnostic algorithm, and evaluating the outcome.
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Topic 2.3: Logic as a Knowledge Representation Framework

1. Discuss the role of logic as a knowledge representation framework in computer science.
 - a) Define logic and explain how it is used to represent facts and rules.
 - b) Discuss the role of logic in reasoning and decision-making, including its use in expert systems and artificial intelligence.
 - c) Construct three original "if-then" statements related to daily school or home life, and explain how each represents a logical rule controlling a decision.
 2. Evaluate why logic-based decision-making is considered more reliable than guesswork, discussing at least two real-world consequences of using inconsistent, non-logical reasoning in a critical system (e.g., medical diagnosis software or airline safety systems).
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Topic 2.4: Propositional Logic

1. Discuss propositional logic in comprehensive detail.
 - a) Define a proposition, and explain why questions and commands cannot be considered propositions.
 - b) Differentiate between simple and compound propositions, providing original examples of each.
 - c) Explain all five logical connectives (AND, OR, NOT, IMPLIES, BICONDITIONAL) in detail, including their symbols and the conditions under which each is true.
 2. Construct a full translation of the following real-world rule into formal propositional logic, showing all steps: "You get free shipping if your order is over \$50 and you are a Premium Member, OR if it is your first purchase." Then test your formula against three different real-world scenarios and determine the outcome for each.
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Topic 2.5: Truth Tables

1. Discuss truth tables in comprehensive detail as a tool for evaluating logical expressions.
 - a) Define a truth table and explain its purpose in propositional logic.
 - b) Describe the step-by-step process of creating a truth table, including how the number of rows is determined for n propositions.
 - c) Differentiate between a tautology, a contradiction, and a contingency, providing an example truth table for each.
 2. Construct a complete truth table for the expression $(P \wedge Q) \vee \neg P$, showing all sub-expression columns. Analyze whether the final expression is a tautology, contradiction, or contingency, and justify your conclusion using the completed table.
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Topic 2.6: Propositional Equivalence

1. Discuss propositional equivalence in comprehensive detail.
 - a) Define propositional equivalence and explain how it is verified using truth tables.
 - b) State and explain De Morgan's two laws, including their historical significance in digital circuit design.
 - c) Construct a full 4-row truth table proving that $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$.
 2. Analyze the practical importance of propositional equivalence in software development and digital circuit design. Discuss how rewriting a security access rule using De Morgan's Law (e.g., "access denied unless valid username AND valid password") could make the logic easier for a programmer to implement.
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Topic 2.7: Propositional Satisfiability

1. Discuss propositional satisfiability in comprehensive detail.
 - a) Define satisfiability and explain the conditions under which a proposition is satisfiable versus unsatisfiable.
 - b) Provide and explain at least two examples each of satisfiable and unsatisfiable expressions.
 - c) Discuss the real-world importance of the SAT problem, including its use in verifying software correctness and safety-critical systems such as autopilot software.
 2. Construct a full truth table to test the satisfiability of the expression $(P \vee Q) \wedge (\neg P \vee \neg Q)$. Evaluate whether the expression is satisfiable, identify all satisfying rows, and explain what real-world logical connective this expression is equivalent to.
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Topic 2.8: Predicate Logic

1. Discuss predicate logic in comprehensive detail as an extension of propositional logic.
 - a) Define predicate logic and explain the limitations of propositional logic that it overcomes.
 - b) Differentiate between propositional logic and predicate logic using a detailed comparison, including the "Ali is Tall" versus "Tall(Ali)" example.
 - c) Explain the roles of predicates and variables in forming predicate logic expressions, with at least three original examples.
 2. Analyze how predicate logic enables more powerful reasoning than propositional logic by translating the following statements into formal predicate logic: "x is greater than 5," "All students are hardworking," and "Some books are interesting." Discuss why propositional logic alone could not adequately express these ideas.
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Topic 2.9: Quantifiers in Predicate Logic

1. Discuss quantifiers in predicate logic in comprehensive detail.
 - a) Define the universal quantifier ($\forall$) and the existential quantifier ($\exists$), including their symbols and meanings.
 - b) Explain, with a detailed example, how combining quantifiers with predicates allows precise logical statements to be written.
 - c) Construct a detailed comparison of " $\forall x \exists y, \text{Loves}(x, y)$ " versus " $\exists y \forall x, \text{Loves}(x, y)$," explaining exactly how swapping quantifier order changes the meaning of the statement.
 2. Design three original real-world statements (not related to the "Loves" example) that each require careful attention to quantifier order to avoid ambiguity. For each statement, provide both possible quantifier orderings and explain the different meaning of each.
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Topic 2.10: Applying Predicate Logic to Real-World Problems

1. Discuss, in comprehensive detail, how predicate logic can be applied to model a real-world system.
 - a) Using the library system example from the chapter, define all relevant predicates (Student, Book, Borrows, Available, HasLibraryCard).
 - b) Construct at least three general rules using quantifiers that describe how the library system should behave.
 - c) Demonstrate, step by step, how a specific fact (e.g., "Student(Ali)") combined with a general rule leads to a specific, reliable conclusion.

2. Design a predicate logic model for a real-world system of your choice (e.g., a school attendance system, a hospital patient system, or a public transport ticketing system). Define at least four predicates, write at least two quantified rules, and demonstrate one full reasoning example showing how a conclusion is drawn from a fact and a rule.
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Topic 2.11: Logic-Based Reasoning and Inference

1. Discuss logic-based reasoning and inference in comprehensive detail.
 - a) Define logical inference and deduction, explaining the relationship between them.
 - b) Explain Modus Ponens, Modus Tollens, and syllogisms in detail, providing a symbolic representation and a real-world example for each.
 - c) Discuss why deduction is considered a reliable method of reasoning, and explain the condition under which its conclusions are guaranteed to be true.
 2. Construct a full step-by-step deductive reasoning trace for a smart home security system, using Modus Tollens, to determine why a specific device did not receive access, based on the rule "If correct password entered (P), then access granted (Q)" and the observed fact that access was not granted. Discuss how this same reasoning pattern could be extended to automatically trigger a security lockout after repeated failures.
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End of Comprehensive Question Bank — Chapter 2: Computational Thinking & Algorithms.